IV Infusions Practice 3

Module 3 - Intravenous Infusions · Daily practice sheet · No date

Name _____ Date _____

a A female patient (35 years old, 65 kg) with normal renal function is given a drug by IV infusion with no loading dose. The elimination half-life is 7 hours. How long after the start of the infusion would it take to reach 95% of the steady-state concentration?

b A female patient (35 years old, 65 kg) is to be given a drug by IV infusion. The elimination half-life is 7 hours and the apparent volume of distribution is 23.1% of body weight. The desired steady-state plasma level is 10 mcg/mL. What is an appropriate infusion rate?

c A female patient (35 years old, 65 kg) with normal renal function is to be given a drug by IV infusion. According to the literature, the elimination half-life of this drug is 7 hours and the apparent V_D is 23.1% of body weight. The pharmacokinetics of this drug assumes a first-order process. The desired steady-state plasma level for this antibiotic is 10 mcg/mL. What is an appropriate loading dose for this antibiotic?

e A female patient (35 years old, 65 kg) with normal renal function is to be given a drug by IV infusion. According to the literature, the elimination half-life of this drug is 7 hours and the apparent V_D is 23.1% of body weight. The pharmacokinetics of this drug assumes a first-order process. The desired steady-state plasma level for this antibiotic is 10 mcg/mL. If the drug were infused for 4 hours at the rate you determined above, what would be the concentration of drug in the plasma at the end of the infusion?

f A female patient (35 years old, 65 kg) with normal renal function is to be given a drug by IV infusion. According to the literature, the elimination half-life of this drug is 7 hours and the apparent V_D is 23.1% of body weight. The pharmacokinetics of this drug assumes a first-order process. The desired steady-state plasma level for this antibiotic is 10 mcg/mL. If the drug were infused for 4 hours at the rate you determined above, what would be the concentration of drug in the plasma 6 hours after the cessation of the infusion?

g A female patient (35 years old, 65 kg) with normal renal function is to be given a drug by IV infusion. According to the literature, the elimination half-life of this drug is 7 hours and the apparent V_D is 23.1% of body weight. The pharmacokinetics of this drug assumes a first-order process. The desired steady-state plasma level for this antibiotic is 10 mcg/mL. If the drug were infused long enough to reach steady-state at the rate you determined above, what would be the concentration of drug in the plasma 8 hours after the cessation of the infusion?

i A female patient (35 years old, 65 kg) with normal renal function is to be given a drug by IV infusion. According to the literature, the elimination half-life of this drug is 7 hours and the apparent V_D is 23.1% of body weight. The pharmacokinetics of this drug assumes a first-order process. The desired steady-state plasma level for this antibiotic is 10 mcg/mL. What rate of infusion and loading dose would you recommend to achieve a steady-state concentration of 30 mg/L? (Answer here: the rate of infusion.)

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Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

a

Her key

Answer: 30.3 hr

WHAT THE STEM GIVES

age	35 years old	*not needed*
weight	65 kg	*not needed*
t½	7 hours	gives $k = \frac{0.693}{7} = 0.099\ \text{hr}^{-1}$
fraction of C_{ss}	95%	so $e^{-kt} = 0.05$ and $kt = -\ln(0.05) = 3.00$

THE WORKING, ONE LINE AT A TIME

SETUP

$0.95 = 1 - e^{-kt}$, so $e^{-kt} = 0.05$

Reaching 95 per cent of steady state means the fraction bracket equals 0.95, leaving 5 per cent of the gap uncovered. The patient's weight and volume of distribution play no part, because the plateau cancels out of a percentage question.

ALGEBRA

$kt = -\ln(0.05) = 3.00$

Taking natural logarithms of both sides gives 2.9957, which she rounds to 3. This is the same reason 3 divided by k is often quoted as a shortcut for the time to 95 per cent of steady state.

ALGEBRA

$$k = \frac{0.693}{7\ \text{hr}} = 0.099\ \text{hr}^{-1}$$

The rate constant is what converts the dimensionless product kt into a time. Dividing 0.693 by 7 hours leaves reciprocal hours.

ALGEBRA

$$t = \frac{3}{0.099\ \text{hr}^{-1}} = 30.3\ \text{hr}$$

A pure number divided by reciprocal hours gives hours. Her second route, 4.32 half-lives multiplied by 7 hours, gives 30.24 hr, the small difference coming from her rounding of 4.32 rather than from any difference of method.

ROUND

$t = 30.3$ hr

Her key prints 30.3 hr, to one decimal place. The answer is a time and carries no reciprocal unit; the age and weight given in the stem never enter it, because the plateau cancels out of any question asked as a percentage of steady state.

> **DOES THE NUMBER MAKE SENSE?**
> 95 per cent of the plateau is reached between four and five half-lives of 7 hours, so t lies between 4 × 7 and 5 × 7 hr; 30.3 hr fits.

b

Her key

Answer: 14.86 mg/hr

WHAT THE STEM GIVES

age	35 years old	*not needed*
weight	65 kg	with the percentage gives $V_D = 0.231 \times 65 =$ 15.015 L
t½	7 hours	gives $k = \frac{0.693}{7} = 0.099\ hr^{-1}$
V_D	23.1% of body weight	the fraction of 65 kg that is the volume
C_{ss}	10 mcg/mL	the target, as 10 mg/L, multiplied by $k \cdot V_D$

THE WORKING, ONE LINE AT A TIME

SETUP

$R = C_{ss} \cdot Cl = C_{ss} \cdot k \cdot V_D$, so a rate constant and an absolute volume both have to be built before the multiplication

The rate that holds a chosen plateau is that plateau times clearance, because at steady state the amount delivered each hour must equal the amount removed. The stem gives a half-life and a percentage of body weight, not a clearance, so those are assembled first, in matching units.

ALGEBRA

$V_D = 23.1\%$ of 65 kg $= 0.231 \times 65 = 15.015$ L

Applying the percentage to the body weight is an evaluation, not a unit conversion. The step from kilograms to litres assumes one kilogram of body weight holds one litre of distribution volume; Dr. Mosley uses this throughout without flagging it, and the answer depends on it entirely.

UNIT

$C_{ss} = 10$ mcg/mL $= 10$ mg/L

Micrograms per millilitre and milligrams per litre are numerically identical, as she instructs the class to take for granted rather than converting. Working in milligrams per litre keeps the final rate in milligrams per hour.

ALGEBRA

$k = \frac{0.693}{7\ hr} = 0.099\ hr^{-1}$

The clearance needed for the rate is the product of the rate constant and the volume of distribution, so the half-life is converted first.

ALGEBRA

$R = C_{ss} \cdot k \cdot V_D = (10\ mg/L)(0.099\ hr^{-1})(15.015\ L) = 14.86\ mg/hr$

The implied clearance is 1.486 L/hr, and 10 multiplied by that gives 14.86. The litres cancel between the volume and the concentration, leaving milligrams per hour.

ROUND

R = 14.86 mg/hr, recommended as 15 mg/hr

Her key prints 14.86 mg/hr and then rounds it to 15 mg/hr as the practical recommendation, writing the two with an arrow between them. Either value falls inside the tolerance here.

DOES THE NUMBER MAKE SENSE?

V_D is 23.1% of 65 kg, 15.015 L, and k is 0.693 over 7, 0.099 hr^{-1}, so Cl is about one and a half L/hr; 10 mg/L times that is about 15 mg/hr, and 14.86 fits.

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c

Her key

Answer: 150 mg

WHAT THE STEM GIVES

R	15 mg/hr	from part b; divided by k
t½	7 hours	gives k = 0.099 hr^{-1}
C_{ss}	10 mcg/mL	10 mg/L, for the $C_{ss} \times V_D$ check
V_D	23.1% of body weight	15.015 L in the 65-kg patient
age	35 years old	not used

THE WORKING, ONE LINE AT A TIME

SETUP

$$D_L = \frac{R}{k}, \text{ or } D_L = C_{ss} \times V_D$$

A loading dose puts the steady-state amount into the body at once. Her key divides the recommended rate by k; the target times the volume gives the same amount.

ALGEBRA

$$k = \frac{0.693}{7 \text{ hr}} = 0.099 \text{ hr}^{-1}$$

0.693 over a time in hours leaves reciprocal hours.

ALGEBRA

$$D_L = \frac{15 \text{ mg/hr}}{0.099 \text{ hr}^{-1}} = 151.52 \text{ mg}$$

mg/hr over hr^{-1}: the hours cancel and mg is left.

ALGEBRA

Check: $D_L = (10 \text{ mg/L})(0.231)(65 \text{ L}) = (10 \text{ mg/L})(15.015 \text{ L}) = 150.15 \text{ mg}$

10 mcg/mL is 10 mg/L, and 23.1% of 65 kg is 15.015 L, taking 1 kg as 1 L. mg/L times L leaves mg.

ROUND

$D_L = 150 \text{ mg}$

Her key prints 151.52 mg → 150 mg, a practical dose; 151.5 mg and 150 mg are both accepted.

DOES THE NUMBER MAKE SENSE?

The loading dose is the amount the body holds at the target: 10 mg/L in 15.015 L is 150.15 mg, and R over k gives 151.52 mg, the same to within rounding.

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e

Her key

Answer: 3.3 mg/L

WHAT THE STEM GIVES

R	15 mg/hr	the numerator of the plateau
Cl_T	1.486 L/hr	the denominator of the plateau
t½	7 hours	gives k for the bracket
t	4 hours	the infusion time in the exponent

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_p = \left(\frac{R}{Cl_T}\right)(1 - e^{-kt}), \text{ with } t = 4 \text{ hr}$$

The infusion runs for 4 hours, well short of 3 to 5 half-lives, so the level at the end is a fraction of the plateau.

ALGEBRA

$$\frac{R}{Cl_T} = \frac{15 \text{ mg/hr}}{1.486 \text{ L/hr}} = 10.094 \text{ mg/L}$$

The plateau the infusion is heading for; mg/hr over L/hr leaves mg/L.

ALGEBRA

$$k = \frac{0.693}{7 \text{ hr}} = 0.099 \text{ hr}^{-1};\ 1 - e^{-(0.099)(4)} = 1 - e^{-0.396} = 0.3270$$

The bracket is the fraction of steady state reached after 4 hours of infusion; kt has no units.

ALGEBRA

$$C_p = (10.094 \text{ mg/L})(0.3270) = 3.30 \text{ mg/L}$$

The plateau times the fraction reached gives the level when the infusion stops.

ROUND

$$C_p = 3.3 \text{ mg/L}$$

Her key prints 3.3 mg/L.

> **DOES THE NUMBER MAKE SENSE?**
> Four hours is a little over half of one 7-hour half-life, so the level is under half of the 10.094 mg/L plateau, about a third of it: 3.3 mg/L fits.

f

Her key

Answer: 1.82 mg/L

WHAT THE STEM GIVES

Cend	3.3 mg/L	the level when the infusion stopped
t½	7 hours	gives k
t	6 hours	time since cessation
infusion time	4-hour	already used to find the 3.3 mg/L

THE WORKING, ONE LINE AT A TIME

SETUP

$C_p = C_{end} \cdot e^{-kt}$, with t counted from cessation

Once the infusion stops only first-order elimination remains, starting from the level at the end of the 4-hour infusion, not from C_{ss}.

ALGEBRA

$$k = \frac{0.693}{7\ \text{hr}} = 0.099\ \text{hr}^{-1}$$

The decay needs the rate constant.

ALGEBRA

$C_p = (3.3\ \text{mg/L})e^{-(0.099)(6)} = (3.3\ \text{mg/L})(0.5521) = 1.82\ \text{mg/L}$

Six hours after cessation is a little under one half-life, so a little over half of 3.3 mg/L remains.

ROUND

$C_p = 1.82\ \text{mg/L}$

Her key prints 1.82 mg/L.

> **DOES THE NUMBER MAKE SENSE?**
> Six hours is a little under one 7-hour half-life, so a little over half of 3.3 mg/L remains: 1.82 mg/L fits.

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g

Her key

Answer: 4.53 mg/L

WHAT THE STEM GIVES

C_{ss}	10 mcg/mL	the level when the infusion stopped
t½	7 hours	gives k
t	8 hours	time since cessation

THE WORKING, ONE LINE AT A TIME

SETUP

$C_p = C_{ss} \cdot e^{-kt}$, with t counted from cessation

This infusion ran to steady state, so the decay starts from C_{ss} itself.

UNIT

$C_{ss} = 10\ \text{mcg/mL} = 10\ \text{mg/L}$

A microgram per millilitre is the same number as a milligram per litre.

ALGEBRA

$C_p = (10\ \text{mg/L})e^{-(0.099)(8)} = (10\ \text{mg/L})(0.4529) = 4.53\ \text{mg/L}$

$k = \frac{0.693}{7\ \text{hr}} = 0.099\ \text{hr}^{-1}$. Eight hours is a little over one half-life, so a little under half remains.

ROUND

$C_p = 4.53\ \text{mg/L}$

Her key starts from the 10 mg/L target. Starting from the 10.09 mg/L that 15 mg/hr gives exactly, the answer is 4.57 mg/L; both are accepted.

DOES THE NUMBER MAKE SENSE?
Eight hours is a little over one 7-hour half-life, so a little under half of 10 mg/L remains: 4.53 mg/L fits.

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i Her key

Answer: 45 mg/hr

WHAT THE STEM GIVES

C_{ss}	30 mg/L	the new target
t½	7 hours	gives k = 0.099 hr^{-1}
V_D	23.1% of body weight	15.015 L for 65 kg
weight	65 kg	with 23.1% gives V_D

THE WORKING, ONE LINE AT A TIME

SETUP

$$R = C_{ss} \times Cl_T = C_{ss} \times k \times V_D$$

At steady state the rate in equals the rate out, $Cl_T \times C_{ss}$, so the rate for a target is the target times clearance.

UNIT

$$V_D = (0.231)(65\ kg) = 15.015\ L$$

23.1% of body weight, taking 1 kg as 1 L.

ALGEBRA

$$k = \frac{0.693}{7\ hr} = 0.099\ hr^{-1};\ Cl_T = (0.099\ hr^{-1})(15.015\ L) = 1.486\ L/hr$$

hr^{-1} times L gives L/hr, the volume of plasma cleared each hour.

ALGEBRA

$$R = (30\ mg/L)(1.486\ L/hr) = 44.6\ mg/hr$$

mg/L times L/hr leaves mg/hr.

ROUND

$$R = 44.6\ mg/hr \rightarrow 45\ mg/hr$$

Her key prints 44.6 mg/hr → 45 mg/hr as the practical rate; both are accepted.

DOES THE NUMBER MAKE SENSE?
Clearance is 1.486 L/hr, a little under one and a half L/hr, and 30 mg/L times that is a little under 45 mg/hr: 44.6 mg/hr, rounded to 45 mg/hr.

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i Her key

Answer: 455 mg

WHAT THE STEM GIVES

R	45 mg/hr	from the rate part; divided by k
t½	7 hours	gives k = 0.099 hr^{-1}
C_{ss}	30 mg/L	with V_D gives the $C_{ss} \times V_D$ check

V_D	23.1% of body weight	15.015 L for 65 kg

THE WORKING, ONE LINE AT A TIME

SETUP

$$D_L = \frac{R}{k}, \text{ or } D_L = C_{ss} \times V_D$$

The loading dose is the amount in the body at the new steady state, given at once.

ALGEBRA

$$D_L = \frac{45 \text{ mg/hr}}{0.099 \text{ hr}^{-1}} = 454.5 \text{ mg}$$

$k = \frac{0.693}{7 \text{ hr}} = 0.099 \text{ hr}^{-1}$; mg/hr over hr^{-1} leaves mg.

ALGEBRA

Check: $D_L = (30 \text{ mg/L})(15.015 \text{ L}) = 450.45 \text{ mg}$

The target times the volume, 23.1% of 65 kg; it differs from 454.5 mg only by the rounding of the rate to 45 mg/hr.

ROUND

$D_L = 454.5 \text{ mg} \rightarrow 455 \text{ mg}$

Her key prints 454.5 mg → 455 mg; a dose near 450 mg is also accepted.

DOES THE NUMBER MAKE SENSE?

30 mg/L in 15.015 L is 450.45 mg, and 45 mg/hr over 0.099 hr^{-1} gives 454.5 mg; 455 mg fits.